

Exercise 2

① Count() function

SELECT count(CID) AS Total Employees
FROM employees;

Total Employees
10

② SELECT SUM(salary) AS Total salary,

Department

FROM employees
~~WHERE~~
GROUP BY department = 'IT';

Total-salary	Department
220 000	IT

③ AVG function

SELECT AVG(salary) AS Average salary,
Department

FROM employees
WHERE Department = 'HR';

Average-salary	Department
49 500	HR

④ MIN and MAX function

SELECT MIN(salary) AS lowest-salary,
MAX(salary) AS Highest-salary
FROM employees;

Lowest - salary	Highest salary
48 000	62 000

⑥ Group BY Statement

```
SELECT sum(salary) as Total salary,
       department
FROM employees
GROUP BY Department;
```

Total salary	Department
220 000	IT
99 000	HR
119 000	finance
105 000	Marketing

⑥ Group BY and COUNT

```
SELECT COUNT(ID) AS Employees, city
FROM employees
GROUP BY City;
```

Employees	City
2	New York
3	Chicago
2	Los Angeles
2	San Francisco
1	Houston

⑦ Group BY and ORDER BY

```
SELECT AVG(salary) AS average salary,
       Department
```


GROUP BY Department
ORDER BY salary DESC;

Average Salary	Department
49 500	HR
24 750 55 000	IT
59 500	Finance
52 500	Marketing

3 HAVING clause

SELECT ~~MAX~~^{SUM}(salary) AS Total salary,
Department
FROM employees
~~WHERE~~^{GROUP BY} Department
HAVING SUM(salary) > 100 000;

Total Salary	Department
220 000	IT
119 000	Finance
105 000	Marketing

9) GROUP BY, HAVING and ORDER BY

SELECT COUNT(~~city~~^{ID}) AS Employees,
City

FROM employees

GROUP BY CITY

HAVING COUNT(ID) > 1

ORDER BY employees DESC;

Employees	City
3	Chicago
2	New York
2	Los Angeles
2	San Francisco

@ Combining Aggregate functions

SELECT AVG (salary) AS Highest Average Salary,
Department

FROM Employees

HAVING AVG (salary) > 'all'

Highest Average Salary	Department
59 500	finance